Punnett squares, peas and crossing

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Punnett

Reginald Crundall Punnett (1875–1967), together with William Bateson, discovered genetic linkage, sex determination, sex linkage and autosomal linkage, using poultry and sweet peas.

The prefix *mono* means 'one' and a *monohybrid cross* is a cross that occurs for a single trait, such as the texture of a pea's outer coat. Mendel observed that the coat can be wrinkled, a recessive trait, or round, a dominant trait. If we consider the wrinkled/round trait, the alleles could be shown as (r) for wrinkled and (R) for round. (The convention is to represent the recessive allele using lower case initial and the dominant allele as an upper case initial). A Punnett square (figure 1) can be drawn to represent a cross between a homozygous round pea (RR) and a homozygous wrinkled pea (rr). The offspring (F_1 generation) are all heterozygous (Rr) peas.

	R	R
r	Rr	Rr
r	Rr	Rr

Figure 1

1 Define *homozygous.*

2 Define *heterozygous.*

3 Use the blank Punnett square (figure 2) to show the genotypes resulting from a cross between (Rr) F_1 offspring.

Figure 2

4 Describe the phenotypes (physical characteristics) of the resulting offspring.

5 What is the ratio of the phenotypes described in Question 4?

6 What is the general name for the offspring of the F_1 generation?

Dihybrid crosses (*di* two) are crosses for two traits, such as the texture *and* colour of the pea's outer coat: wrinkled (r) or round (R) and yellow (g) or green (G). Dihybrid Punnett squares are done in much the same way as monohybrid crosses, except that you are working with two traits. Imagine a cross between seeds with a genotype of RrGg. The possible gametes are RG, Rg, rG and rg. Figure 3 shows all the genotypes for the RrGg x RrGg cross.

7 In the space below each cross in figure 3, write the resulting phenotype for each individual genotype.

	RG	Rg	rG	rg
RG	RRGG	RRGg	RrGG	RrGg
Rg	RRGg	Rrgg	RrGg	Rrgg
rG	RRGG	RrGg	rrGG	rrGG
rg	RrGg	Rrgg	rrGg	rrgg

Figure 3

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